

PRAKTIKUM SISTEM OPERASI

MODUL V

EKSPLORASI PROSES XINU



Disusun Oleh :

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PROGRAM STUDI S1 SOFTWARE ENGINEERING

FAKULTAS INFORMATIKA

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JURNAL

1. [10 Poin] Jawablah pertanyaan berikut ini:

a. Berapa banyaknya maksimum proses yang ada pada Xinu?

Jawaban: Maksimum jumlah proses pada Xinu adalah 8 proses, yang didefinisikan dengan konstanta NPROC = 8

b. Berapa maksimal panjang nama suatu proses pada Xinu?

Jawaban: 16 karakter define PNMLEN 16 termasuk karakter akhir (\0)

c. Berapa nilai prioritas awal pada saat proses dibuat?

Jawaban: 20 (Fungsi create() secara default menetapkan prioritas PR_DEFAULT = 20)

d. Ada berapa total state pada Xinu? Sebutkan!

Jawaban: 7 state

Yaitu:

1. PR_FREE: proses kosong / tidak digunakan
2. PR_CURR: sedang berjalan (running)
3. PR_READY: siap dijalankan
4. PR_RECV: menunggu pesan
5. PR_SLEEP: sedang tidur
6. PR_SUSP: ditangguhkan
7. PR_WAIT: menunggu semaphore

2. [20 Poin] Perintah ps adalah perintah untuk menampilkan statistik process yang berjalan.

Source code dari ps tersimpan pada file xsh_ps.c. Carilah file tersebut dan beri komentar pada 20 baris terakhir di source code tersebut!

Jawaban:

```
Activities Terminal Sat 11 Apr, 08:04
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help
xinu@xinu-develop-end:~$ find / -name "xsh_.c" 2>/dev/null
xinu@xinu-develop-end:~$ ls
prev-version  README  xinu
xinu@xinu-develop-end:~$ cd xinu
xinu@xinu-develop-end:~/xinu$ ls
banner.h  config  DESCRIPTION  include  main.c  shell
compile  COPYRIGHT  device  lib  net  system
xinu@xinu-develop-end:~/xinu$ find . -name "xsh_ps.c"
./shell/xsh_ps.c
xinu@xinu-develop-end:~/xinu$ cat -n ./shell/xsh_ps.c
1  /* xsh_ps.c - xsh_ps */
2
3  #include <xinu.h>
4  #include <stdio.h>
5  #include <string.h>
6
7  /*-----
8   * xsh_ps - shell command to print the process table
9   *-----
10  */
11  shellcmd xsh_ps(int nargs, char *args[])
12  {
13      struct proctent *prptr;          /* Pointer to process */
14      int32 i;                        /* Index into proctabl */
15      char *pstate[] = {               /* Names for process states */
16          "free ", "curr ", "ready", "recv ", "sleep", "susp ",
17          "wait ", "rtime"};
18
```

```
Activities Terminal Sat 11 Apr, 08:04
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help
19  /* For argument '--help', emit help about the 'ps' command */
20
21  if (nargs == 2 && strcmp(args[1], "--help", 7) == 0) {
22      printf("Use: %s\n\n", args[0]);
23      printf("Description:\n");
24      printf("\tDisplays information about running processes\n");
25      printf("Options:\n");
26      printf("\t--help\t display this help and exit\n");
27      return 0;
28  }
29
30  /* Check for valid number of arguments */
31
32  if (nargs > 1) {
33      fprintf(stderr, "%s: too many arguments\n", args[0]);
34      fprintf(stderr, "Try '%s --help' for more information\n",
35              args[0]);
36      return 1;
37  }
38
39  /* Print header for items from the process table */
40
41  printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
42         "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
43         "Stack Ptr", "Stack Size");
44
45  printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
46         "---", "-----", "-----", "-----", "-----", "-----",

```

```
Activities Terminal Sat 11 Apr, 08:05
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help
36         return 1;
37     }
38
39     /* Print header for items from the process table */
40
41     printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
42           "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
43           "Stack Ptr", "Stack Size");
44
45     printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
46           "----", "-----", "-----", "-----", "-----",
47           "-----", "-----", "-----");
48
49     /* Output information for each process */
50
51     for (i = 0; i < NPROC; i++) {
52         prptr = &proctab[i];
53         if (prptr->prstate == PR_FREE) { /* Skip unused slots */
54             continue;
55         }
56         printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d\n",
57               i, prptr->prname, pstate[(int)prptr->prstate],
58               prptr->prprio, prptr->prparent, prptr->prstkbase,
59               prptr->prstkptr, prptr->prstklen);
60     }
61     return 0;
62 }
xinu@xinu-develop-end:~/xinu$
```

```
Activities Terminal Sat 11 Apr, 08:31
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help
GNU nano 2.2.6 File: ./shell/xsh ps.c Modified

}

/* Print header for items from the process table */

printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
       "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
       "Stack Ptr", "Stack Size");

printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
       "----", "-----", "-----", "-----", "-----",
       "-----", "-----", "-----");

/* Output information for each process */

for (i = 0; i < NPROC; i++) {
    prptr = &proctab[i];
    if (prptr->prstate == PR_FREE) { /* Skip unused slots */
        continue;
    }
    printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d\n",
          i, prptr->prname, pstate[(int)prptr->prstate],
          prptr->prprio, prptr->prparent, prptr->prstkbase,
          prptr->prstkptr, prptr->prstklen);
}

^G Get Help ^O WriteOut ^R Read File ^Y Prev Page ^K Cut Text ^C Cur Pos
^X Exit ^J Justify ^W Where Is ^V Next Page ^U UnCut Text ^T To Spell
```

The image displays two screenshots of a terminal window, showing the modification of the `xsh_ps.c` file in the Xinu operating system. The terminal window title is `xinu@xinu-develop-end: ~/xinu`.

Top Screenshot (Sun 12 Apr, 08:22): Shows the initial code in the `xsh_ps.c` file. The code includes a `printf` statement for process information, with columns for `Pid`, `Name`, `State`, `Prio`, `Ppid`, `Stack Base`, `Stack Ptr`, `Stack Size`, `Msg`, and `Content`. The code is written in C, using `prptr` and `prstate` to access process information.

```

}

/* Print header for items from the process table */
/* Menampilkan header/judul kolom untuk tabel proses*/

printf("%3s %-16s %5s %4s %4s %10s %-10s %10s %3s %8s%05\n",
       "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
       "Stack Ptr", "Stack Size", "Msg", "Content");

printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
       "----", "-----", "-----", "-----", "-----", "-----",
       "-----", "-----", "-----", "-----");

/* Output information for each process */
/*Alya Rabani - 2311104076 - Jurnal 5*/

for (i = 0; i < NPROC; i++) {
    prptr = &proctab[i];
    if (prptr->prstate == PR_FREE) { /* Skip unused slots */
        continue;
    }
    printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d %3d %8d\n",
           i, prptr->prname, prstate[(int)prptr->prstate],
           prptr->prprio, prptr->prparent, prptr->prstkbase,

```

Bottom Screenshot (Sat 11 Apr, 08:45): Shows the modified code. The `printf` statement has been updated to include an additional column for `prstkptr` and `prstklen`. The code is written in C, using `prptr` and `prstate` to access process information.

```

/* Print header for items from the process table */
/* Menampilkan header/judul kolom untuk tabel proses*/

printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
       "Pid", "Name", "State", "Pprio", "Ppid", "Stack Base",
       "Stack Ptr", "Stack Size");

printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
       "----", "-----", "-----", "-----", "-----", "-----",
       "-----", "-----", "-----", "-----");

/* Output information for each process */
/*Alya Rabani - 2311104076 - Jurnal 5*/

for (i = 0; i < NPROC; i++) {
    prptr = &proctab[i];
    if (prptr->prstate == PR_FREE) { /* Skip unused slots */
        continue;
    }
    printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d\n",
           i, prptr->prname, prstate[(int)prptr->prstate],
           prptr->prprio, prptr->prparent, prptr->prstkbase,
           prptr->prstkptr, prptr->prstklen);
}
return 0;
}
xinu@xinu-develop-end:~/xinu$

```

3. [35 Poin] Ubahlah perintah `ps` (source code: `xsh_ps.c`) pada Xinu sehingga menampilkan informasi tambahan berupa kolom yang berisi total message yang ada pada proses seperti gambar di bawah ini:

```
xsh $ ps
```

Pid	Name	State	Prio	Ppid	Stack Base	Stack Ptr	Stack Size	Msg	Content
0	prnull	ready	0	0	0x005FDFFC	0x00FFFF04	8192	1	4
1	rdsproc	susp	200	0	0x005FBFFC	0x005FBFC8	16384	0	0
2	ipout	wait	500	0	0x005F7FFC	0x005F7F40	8192	0	0
3	netin	wait	500	0	0x005F5FFC	0x005F5EE0	8192	0	0
5	Main process	recv	20	4	0x005E3FFC	0x005E3F64	65536	0	0
6	shell	recv	50	5	0x005D3FFC	0x005D3C7C	8192	0	0
7	ps	curr	20	6	0x005F3FFC	0x005F3DC8	8192	0	0

Kolom Msg adalah banyaknya pesan yang ada dalam proses.

Kolom Content adalah isi dari pesan tersebut.

Langkah pengerjaan:

- Modifikasi source code pada file xsh_ps.c

```

}

/* Print header for items from the process table */
/* Menampilkan header/judul kolom untuk tabel proses*/

printf("%3s %-16s %5s %4s %4s %10s %-10s %10s %3s %8s\n",
       "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
       "Stack Ptr", "Stack Size", "Msg", "Content");

printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
       "----", "-----", "-----", "-----", "-----",
       "-----", "-----", "-----", "-----");

/* Output information for each process */
/*Alya Rabani - 2311104076 - Jurnal 5*/

for (i = 0; i < NPROC; i++) {
    prptr = &proctab[i];
    if (prptr->prstate == PR_FREE) { /* Skip unused slots */
        continue;
    }
    printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d %3d %8d\n",
           i, prptr->prname, pstate[(int)prptr->prstate],
           prptr->prprio, prptr->prparent, prptr->prstkbase,

```

- Kompilasi ulang Xinu dengan perintah seperti pada modul sebelumnya
- Jalankan Backend VM
- Setelah sistem berjalan, jalankan perintah \$ps. Pastikan hasilnya sesuai dengan contoh output pada gambar yang diberikan.

```
Activities Terminal Tue 14 Apr, 00:21
xinu@xinu-develop-end: ~/xinu/compile

-----ALYA RABANI-----
XINU
-----2311104076-----

Welcome to Xinu!

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size Msg  Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFE4  8192  1    3
1 ipout           wait   500  0 0x005FBFFC 0x005FBF30  8192  0    0
2 netin           wait   500  0 0x005F9FFC 0x005F9ED0  8192  0    0
4 Main process    recv   20   3 0x005E7FFC 0x005E7F34 65536  0    0
5 shell           recv   50   4 0x005F7FFC 0x005F79AC  8192  0    0
6 ps              curr   20   5 0x005F5FFC 0x005F5FC4  8192  0    0

xsh $

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0
```

- Screenshot source code dan output akhir hasil modifikasi

4. **[35 Poin]** Ubahlah perintah uptime pada Xinu sehingga menampilkan lamanya Xinu sejak booting hanya dalam satuan menit.

Langkah pengerjaan:

- Modifikasi source code pada file xsh_uptime.c
- Kompilasi ulang Xinu dengan perintah seperti pada modul sebelumnya
- Jalankan Backend VM
- Setelah sistem berjalan, jalankan perintah \$uptime. Pastikan hasilnya sesuai dengan contoh output yang diinginkan
- Screenshot source code dan output akhir hasil modifikasi

```
Activities Terminal Tue 14 Apr, 00:42
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help
xinu@xinu-develop-end:~$ cd xinu
xinu@xinu-develop-end:~/xinu$ find . -name "xsh_uptime.c"
./shell/xsh_uptime.c
xinu@xinu-develop-end:~/xinu$ cat -n ./shell/xsh_uptime.c
cat: ./shell/xsh_uptime.c: No such file or directory
xinu@xinu-develop-end:~/xinu$ cat -n ./shell/xsh_uptime.c
 1 /* xsh_uptime.c - xsh_uptime */
 2
 3 #include <xinu.h>
 4 #include <stdio.h>
 5 #include <string.h>
 6
 7 /*-----
 8  * xsh_uptime - shell to print the time the system has been up
 9  *-----
10  */
11 shellcmd xsh_uptime(int nargs, char *args[])
12 {
13     uint32  days, hrs, mins, secs; /* Days, hours, minutes, and */
14                                     /* seconds since system boot */
15     uint32  secpday = 86400;        /* Seconds in a day */
16     uint32  secperrh = 3600;        /* Seconds in an hour */
17     uint32  secperrm = 60;         /* Seconds in a minute */
18
19     /* For argument '--help', emit help about the 'uptime' command */
20
21     if (nargs == 2 && strcmp(args[1], "--help", 7) == 0) {
22         printf("Use: %s\n\n", args[0]);
```

```
Activities Terminal Tue 14 Apr, 00:48
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help
GNU nano 2.2.6 File: ./shell/xsh_uptime.c Modified

/* xsh_uptime.c - xsh_uptime */

#include <xinu.h>
#include <stdio.h>
#include <string.h>

/*-----
 * xsh_uptime - shell to print the time the system has been up
 *-----
 */
shellcmd xsh_uptime(int nargs, char *args[])
{
    uint32  days, hrs, mins, secs; /* Days, hours, minutes, and */
                                    /* seconds since system boot */
    uint32  secpday = 86400;        /* Seconds in a day */
    uint32  secperrh = 3600;        /* Seconds in an hour */
    uint32  secperrm = 60;         /* Seconds in a minute */

    /* For argument '--help', emit help about the 'uptime' command */

    if (nargs == 2 && strcmp(args[1], "--help", 7) == 0) {
        printf("Use: %s\n\n", args[0]);
        printf("Description:\n");
    }
}
```

^G Get Help ^O WriteOut ^R Read File ^Y Prev Page ^K Cut Text ^C Cur Pos
^X Exit ^J Justify ^W Where Is ^V Next Page ^U UnCut Text ^T To Spell


```
Activities Terminal Tue 14 Apr, 00:56
xinu@xinu-develop-end: ~/xinu
File Edit View Search Terminal Help
xinu@xinu-develop-end:~/xinu$ cat -n ./shell/xsh_uptime.c
1  /* xsh_uptime.c - xsh_uptime */
2
3  #include <xinu.h>
4  #include <stdio.h>
5  #include <string.h>
6
7  /*-----
8   * xsh_uptime - shell to print the time the system has been up
9   *-----
10 */
11 /* Alya Rabani - 2311104076 - Jurnal 5 */
12 shellcmd xsh_uptime(int nargs, char *args[])
13 {
14     uint32  days, hrs, mins, secs; /* Days, hours, minutes, and */
15                                     /* seconds since system boot */
16     uint32  secperday = 86400;      /* Seconds in a day */
17     uint32  secperhr  = 3600;       /* Seconds in an hour */
18     uint32  secpermin  = 60;        /* Seconds in a minute */
19
20     /* For argument '--help', emit help about the 'uptime' command */
21
```